Remittances, Child Labor, and Schooling: Evidence from Colombia*

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Abstract

We use a nationally-representative household dataset from Colombia to study the impact of remittances on child labor. As the source of identification, we leverage the unemployment shocks in the main destination countries of Colombian migrants arising from the 2008 financial crisis and combine them with information on the historical migration rates from Colombian regions to these countries. We estimate the impact of remittances on child labor at both the extensive and intensive margins, where we account for the zero-inflated nature of hours worked for the latter. We find that remittances reduce both child labor participation and hours worked. Our heterogeneous analysis suggests that the impact primarily affects girls, younger children, and children living in non-poor households. Moreover, we also find that remittance income has no effect on schooling participation.

Keywords: Child Labor, Schooling, Remittances, Colombia
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1 Introduction

An estimated 168 million, or about 11 percent of children worldwide are child laborers (ILO, 2013).[1] Many of these children, given household budget constraints, often work out of necessity (IPEC, 2013). Yet, their need to fulfill current consumption may have negative intertemporal consequences. Work may harm the human capital accumulation (Basu and Van, 1998; Baland and Robinson, 2000), learning outcomes (Emerson et al., 2016), the health (O'Donnell et al., 2005), and future labor market outcomes (Emerson and Souza, 2011) of these children.

Interventions that relax budget constraints can reduce child labor (Basu and Van, 1998). In some developing countries, policy makers use public resources to intervene through government transfer programs (see e.g., Attanasio et al., 2010; De Carvalho, 2012; Galiani and McEwan, 2013). However, in countries that send international migrants, private remittance income can play a similar function for households with emigrating members (Calero et al., 2009; Acosta, 2011; Alcaraz et al., 2012; Bargain and Boutin, 2015). Similar to cash transfers, remittance income can potentially reduce child labor in economies with a significant share of international migrants (de Hoop and Rosati, 2014b).

In this paper, we study the impact of remittances on child labor in Colombia. About 12 percent of Colombian children and teens aged 12-18 living in Colombia's major metropolitan areas work. At the same time, international migrants are important contributors to the Colombian economy. In absolute terms, Colombia is the fourth largest recipient of remittance income in Latin America and among the top 30 largest recipients in the world (Yang, 2011). Yet, despite its importance, little is known about the effect of remittances on child labor. We develop a strategy to identify the causal effect of remittances on child labor. To our knowledge, this is the first study of the impact of remittances on child labor in Colombia.

Our analysis uses repeated cross-sections of the household survey of Colombia's major cities and their metropolitan areas, *Gran Encuesta Integrada de Hogares* (GEIH). This dataset includes information on children's labor participation and working hours, which allows us to estimate the impact of remittances at the extensive and intensive margins. Previous studies that look at the question tend to focus on the former.[2] This focus may overlook the impact on households whose remittance income was not large enough to substitute for their children's income. Moreover, conditional on participation, child labor affects human capital accumulation primarily by diverting potential hours away from learning.

To identify the impact of remittances, we focus our analysis on the 2008–2010 period and estimate an instrumental variable (IV) model. For the IV, we leverage the employment shocks

[1] There are estimated to be around 78 million child workers in Asia and the Pacific, 59 million in Sub-Saharan Africa, and 13 million in Latin America and the Caribbean.

[2] More recently, Dimova et al. (2015) is an exception, albeit with a different focus on agricultural child labor in rural Tanzania. Other related papers that look at the intensive margin of child labor use the migration status as the variable of interest (e.g., Yang, 2008; Antman, 2011; Mansuri, 2006).

experienced by Colombia's most important migrant destinations that arose due to the 2007/2008 financial crisis. We focus on the immediate years after the crisis to minimize the effect of the households' endogenous adjustment that would have occurred over a longer time horizon (Yang, 2008).

We construct the IV by weighting the unemployment shocks — defined as deviations from the pre-crisis unemployment rate — in Colombian migrants' main destination countries (i.e., the United States, Spain, and Venezuela) for each respective year with the historical share of migrants to these countries for each sending region in the data. To address concerns over regional confounders, we include a set of regional controls. Moreover, with an IV that varies across regions *and* over time, we can further address concerns over regional confounders using region fixed effects for our extensive margin estimates.[3]

We argue that this variable is an appropriate IV for remittances received by Colombian households. Macroeconomic shocks (as proxied by the unemployment shocks) in a destination country will affect the income of migrants who reside there, which in turn will affect the remittances received by the households left behind.[4] Indeed, our regression finds a negative correlation between the migrant-share-weighted average of the unemployment shocks in the destination countries and the remittance receipts of Colombian households. At the same time, in the short term, these are unanticipated shocks for recipient households. Hence, conditional on the regional control variables — and, in the case of the extensive margin estimates, the region fixed effects — these unemployment shocks in the destination countries should be excludable from the (second stage) child labor regression.

We find that remittances reduce child labor. At the extensive margin, our preferred estimates suggest that a (PPP-adjusted, which hereafter is implied) US$1 increase in the remittance income received by households decreases the incidence of child labor by 0.017 percentage points (p.p.). This suggests that a 10-percent increase at the mean remittance value of US$ 4,040 leads to a 6.8 p.p. reduction in the probability of child labor. Moreover, we find that at the intensive margin, a 10-percent increase at the mean remittances reduces the number of hours worked per week by 2.3 hours, which is equal to 6 percent of the average hours worked.

We disaggregate the effect by gender and age. In terms of gender, we find that the effect is only statistically significant among girls at both the extensive and intensive margins. At the same time, we do not find a meaningful difference in the impact between younger (12–14 years old) and older (15–19 years old) children. Remittances reduce child labor in all age groups, with similar point estimates at both the extensive and intensive margins. However, at the intensive margin, it is only statistically significant for younger children.

[3] As we explain in Section 5, our preferred specification for the intensive margin estimates does not include the region fixed effects given consistency issues associated with the inclusion of fixed effects in nonlinear (Tobit) estimations.

[4] A number of previous studies used macroeconomic shocks to identify the impact of remittances on the welfare of households left behind (see, e.g. Yang, 2008; Alcaraz et al., 2012). The study most similar to ours is by Amuedo-Dorantes and Pozo (2010), who use unemployment rates and average real service sector earnings in the United States as an IV for whether households receive remittances.

To understand whether the effect is heterogeneous by economic status, we disaggregate the impact of remittances by the household wealth. To capture household wealth, we construct an asset-based wealth index for each household. Based on this index, we define two wealth groups: poor (below the median asset index) and non-poor (above the median asset index) households. Remittance impacts are negative among both types of households. However, we find that the impacts are only statistically significant for non-poor households, consistent with the notion that these are the households with higher reservation wages.

Finally, we explore whether remittances increase school participation. One key policy issue related to child labor is its potential negative impact on schooling. For instance, the 2012 National Schooling Desertion report suggests that work is one of the main causes of primary and secondary school drop-outs in the Caribbean and Pacific regions (MEN, 2012). It is therefore plausible that remittances would impact school participation in a similar (albeit opposite) way to child labor by relaxing budget constraints. However, we do not find that remittances affect school participation in Colombia. This finding may be driven by the fact that many children in Colombia often combine working and attending school (Scott et al., 2016). It is also consistent with broader evidence that being out of the labor market does not necessarily imply being in school (and vice versa) (Edmonds and Pavcnik, 2005).

The rest of the paper is organized as follows: Section 2 presents a backdrop for our analysis by describing migration and child labor in Colombia. Section 3 presents a simple model that guides the interpretation of the results in the paper. Section 4 provides a description of the data and the variables. Section 5 describes the empirical strategy. Section 6 and 7 report the main results and the heterogeneous impact analysis. Section 8 reports some results in terms of schooling. Section 9 concludes.

2 International Migration and Child Labor in Colombia

In this section, we provide the institutional context for the analysis. We describe the historical pattern of international migration from Colombia and its effects in terms of remittance flows. We then discuss the regulatory framework on child labor and present evidence of its enforcement in Colombia.

2.1 Migration and Remittances

International migration has played an important role in Colombia's economic development in the last few decades (Cardona-Sosa and Medina, 2006). The Colombian statistical agency, *Departamento Administrativo Nacional de Estadística* (DANE), estimated that, by 2005, around 8 percent of the total population lived outside the country, and a bulk of this migration was motivated by the desire to look for better economic and job opportunities (ENMIR, 2009). Historically, the main destinations

included the United States, Spain, and Venezuela (OIM, 2010).[5] Over time, the United States and Spain increasingly became the most preferred destinations: Between 2000 and 2010, the number of migrants to the United States almost doubled (from 471 to 909 thousand) while that to Spain increased more than tenfold (from 36 to 373 thousand) (OIM, 2012).

Figure 1 shows the historical net migration rate across the different regions in Colombia.[6] It illustrates the wide spatial distribution of migrants hailing from these regions. Over time, they developed migrant networks in their destination countries, and migrants from a particular region often have specific ties to networks in a particular country. Migrants from the coffee region in the center of the country prefer Spain over other destinations. Those from the main cities — Bogotá, Medellín and Cali — prefer the United States as their main destination. Meanwhile, migrants from the northwest region prefer to its bordering country of Venezuela.

The rapid growth of outmigration brought a significant inflow of remittances into Colombia. According to the central bank *Banco de la República*, remittance flows increased from US$1,578 millions in 2000 to US$3,313 millions in 2005 and reached a peak of US$4,427 millions in 2008. In 2008, an overwhelming 73 percent of the remittances came from two countries: the United States (37 percent) and Spain (36 percent). However, the 2008 financial crisis that began in the United States and spread to the Eurozone led to the slowing down of remittance receipts. It was only by 2015 that the remittance inflow returned to the pre-crisis level.

For the households left behind, about 59 percent of the remittance income is used for recurrent household expenditure (Garay and Rodriguez, 2005). Moreover, only 4 percent of this income goes into household savings. Hence, remittances function as a means to relax household immediate budget constraints and have an important role in the survival of many poor households. Since helping out with household expenses account is an important reason why children work in Colombia, remittance income may reduce child labor through its effect on household budget constraints.

2.2 Child Labor

The Government of Colombia regulates the allowed working hours and sectors for working children of different ages. According to Law 1098, children less than 15 years old can only work for a maximum of 14 hours during the week, and only in artistic, cultural, recreational or sports activities. Meanwhile, children between 15 and 16 years old require authorization to work a maximum of 30 hours during the week. Those between 17 and 18 years old can work without restriction but for no more than 40 hours in a week. Interestingly, the *other services* sector, which

[5]Based on the 2005 census, DANE estimated that the majority of Colombia's international migrants worked in three countries: Venezuela (20 percent), Spain (23.1 percent), and the United States (34.6 percent) (OIM, 2012). By 2010, these countries still account for more than 70 percent of these migrants (World Bank, 2011).

[6]Data on regional migration is constructed by the National Department of Statistics (DANE) and aims to provide adjusted population data from the 2005 general census. For the migration component, the analysis includes information provided by the population censuses, which allows us to construct historical net migration rates.

includes artistic, cultural, and recreational activities only employs 2.3 percent of the total working children between 12 and 14 years old, despite the fact that the law allows these children to work only in those activities.

According to the 2008–2010 GEIH data, around 12 percent of Colombian children work and there are important differences in the types of work they undertake by gender. Table 1 shows total participation by main economic activity. The main sectors that employ children are wholesale and retail, followed by hotels and restaurants, manufacturing (mainly of food and beverage products), and storage and transportation (which includes mail delivery and communications). Across gender, the wholesale and retail sector is the primary employer. However, for boys, the construction sector is the second-most important employer, employing 8.9 percent of working boys, compared to 0.3 percent for girls. For girls, the domestic service sector is the second-most important employer that employed 13 percent working girls, compared to a mere 0.3 percent for boys.[7]

Our data suggest that the age restriction regulation has not been strictly enforced. In Table 2, we show hours worked by age groups (as stipulated in the law) and gender. Children aged 12-14 years old not only work in activities that are different to what is legally allowed, but also report working more hours than the legal limit. On average, children with 12-14 years old work more than 20 hours during the week with boys reporting working more hours than girls. Importantly, we find that the average number of working hours exceeds the legal limits, but only for boys. The number of hours worked exceeds the 30 hours limit for boys with 15-16 years old and the 40 hours limit for boys with 17-18 years old.[8] These numbers are consistent with qualitative evidence suggesting a stronger expectation for boys to be self sufficient (Pinzón-Rondón et al., 2008).

3 Conceptual Framework

We follow the standard neoclassical framework to examine how remittances affect child labor (Killingsworth et al., 1986). Households maximize a utility function based on household consumption and child leisure, $U(C, L)$, subject to the budget constraint of the household. The model assumes that parents send their children to work based on the value of child wage contribution and living standards (Edmonds, 2007). We assume that the living standard of the household is given by household consumption, C, which is equal to the sum of the household's exogenous income, Y, plus revenue from child labor multiplied by the time children spend working, wH, plus the amount of remittances the household receives, R. Moreover, we assume that total time available for child labor, H, and total time for leisure, L, is normalized to 1.

[7] Domestic service refers only to paid work and includes cleaning, gardening, baby-sitting, cooking, driving, and other housework.

[8] It is important to note that the GEIH includes information of the number of working children in the thirteen main cities of Colombia and their metropolitan areas, but data are not collected for hard-to-reach populations, including children involved in illicit activities.

The household objective function is to choose consumption and child labor supply that maximize the household utility subject to the budget constraint of the household:

$$Max_{C,L}U(C,L)$$

$$s.t$$

$$C = Y + wH + R$$

Thus, households solve the following Lagrangean equation:

$$\mathcal{L} = U(C,L) + \lambda(Y - C + wH + R)$$

From the first order conditions

$$\frac{\partial \mathcal{L}}{\partial C} = U'_C(C,L) - \lambda = 0$$
$$\frac{\partial \mathcal{L}}{\partial L} = U'_L(C,L) - w\lambda = 0$$

The equilibrium child labor is determined by equalizing the marginal rate of substitution between consumption and leisure (U'_L/U'_C) to the wage rate w. If $(U'_L/U'_C) = w$ at $H = 0$, there is a corner solution and the child does not work. At this point, market wage equals to the reservation wage and parents are indifferent about sending children to work or not.

Figure 2 illustrates the predictions of the model. Remittances supplement households' non-labor income, relax household budget constraints, and reduce child labor (Basu and Van, 1998; de Hoop and Rosati, 2014b). For some households, the remittances would only be enough to affect child labor at the intensive margin, but not at the extensive margin (i.e., the move from A to B). With large enough remittances, there will be an effect at both margins, as households no longer need to have their children work (i.e., the corner solution, C).

4 Data and Measurements

To account for the impact of total amount of remittances in the household on child labor, we use the Colombian household survey *Gran Encuesta Integrada de Hogares* (GEIH). This survey is collected by DANE and includes socio-economic characteristics of the population such as labor, gender, age, and income sources. It is the largest national survey in terms of coverage, and includes information of the thirteen main cities and their metropolitan areas. We pool the cross-section data for the 2008–2010 period to obtain observations for 146,773 children and teens

between 12 and 18 years old.[9]

Table 3 displays the summary statistics of children and household characteristics. Of the 146,773 children, 5,713 live in remittance-receiving households. Those households have a higher proportion of female household head, which may be related to migration patterns. The proportion of children working and the number of hours worked are higher in non-recipients. Also, the proportion of employed household head is higher in non-recipient households, which suggests that remittances may be used to buy time away from work by the household head.

4.1 Child Labor and Remittances

We estimate the effect of remittances on children's labor participation for children aged 12–18 years old. We use two variables to capture labor participation at the extensive and intensive margins. First, we use a binary variable of whether the child reports participating in any type of economic activities as the dependent variable. The list of economic activities includes wage labor, family labor, domestic work, and unpaid work. Then, we use the question on the typical number of hours the child works to estimate the impact at the intensive margin.

Our main regressor is the total amount of international remittances received by the household in the last twelve months. *GEIH* asked each individual household member for the amount of remittances received from persons residing outside of the country. We construct the remittance variable by summing up these values for each household. We then convert this variable to the PPP-adjusted US dollar equivalent.[10] In discussions using our data, all references to US$ are PPP-adjusted.

4.2 Other Observables

We include two sets of control variables at the household and regional levels. First, we include a set of control variables that capture child and household (and household head) characteristics. For the former, we include the gender and age of the child. The set of household characteristics includes the household size, the number of children in the household, the education of the household head, and three dummy variables for the gender, marital status, and employment status of the household head.

Meanwhile, we use household assets instead of current income as a proxy of living standard, to address concerns over the potential endogeneity of income (McKenzie, 2005). We use the principal component analysis to construct this asset index. The variables used to construct this index include: house ownership; the availability of electricity, telephone, and internet services;

[9]The survey includes information for a cohort of 49,212 children between 12 and 18 years old in 2008; 48,925 in 2009; and 48,636 in 2010.

[10]We use the PPP conversion factor for private consumption from the World Bank's *World Development Indicators* database (World Bank, 2016).

sources of water supply; sanitation facility; and the types of flooring material. Table A.1 in the Appendix provides the full list of the variables used to construct this index.

The asset index allows us to explore the variation in child labor participation in households with different socio-economic status (Huang, 2013). While the proportion of children from the richest quartile in the labor force is 8 percent, in the poorest quartile this proportion reaches almost 16 percent. In terms of hours worked, children living in the richest quartile households also report working less hours. For the poorest quartile, the average number of hours worked is 36 hours during the week, while those children from households at the richest quartile report working on average 31 hours during the week.

To account for region-specific economic characteristics, we include the Gini coefficient and the unemployment rate for each year in the analysis to capture regional income inequality and the labor market equilibrium. Moreover, given that we do not include the region fixed effects in the intensive margin analysis (for reasons explained in Section 6.2), we control for the differential strength of the migrant networks in the regions using the historical net migration rate between 2000–2005. All of the regional-level variables are obtained from DANE.

5 Empirical Strategy

We estimate the impact of remittances on participation and hours worked. For the participation variable, we first estimate the following linear probability model:

$$Y_{iht} = \gamma R_{ht} + \mathbf{X_{it}}\beta + \varepsilon_{it}$$

where Y_{iht} is an indicator variable that equals one if child i during month t is working. R_{ht} is the variable of interest, the total amount of remittances that household h received in the twelve months prior to time t. $\mathbf{X_{it}}$ is a vector of household and child characteristics. The coefficient of interest, γ, is the estimated effect of an additional unit of remittances in the incidence of child labor.

Meanwhile, to estimate the remittance impact on work hours, we begin by addressing the censored nature of working hours. We use the standard approach by estimating the Tobit regression of work hours on remittances:

$$Y_{iht} = \alpha R_{ht} + \mathbf{X_{it}}\alpha + \varepsilon_{it}$$

$$\varepsilon_{it} \sim Normal(0, \sigma^2)$$

$$Y_{iht} = Max(0, Y_i^*)$$

where R_{ht} is our main independent variable, and $\mathbf{X_{it}}$ is the same vector of explanatory variables

as before.

Remittance income is potentially endogenous to child labor outcomes. Unobservable idiosyncratic shocks to household income, for instance, may influence both the amount of remittances received and child labor outcomes. We address this problem by estimating an instrumental variable (IV) model. To identify the causal impact of remittances, we leverage the sharp unemployment shocks experienced by a number of Colombia's main migration destination countries arising from the 2007/2008 financial crisis in the United States and, subsequently, in the Eurozone. The crisis led to an economic recession and a sharp increase in unemployment in both the United States and Spain.[11] We use the deviation from the average pre-crisis unemployment rate as a component of our IV. Figure 3 shows the link between unemployment rates in destination countries and remittances from those countries: an increase in unemployment at each destination is associated with a lower remittance flow from that country.

We interact the unemployment shocks with the historical rate of migration from each region to the three main destination countries, the United States, Spain, and Venezuela. Empirical evidence suggests that regional migration patterns indicate the strength of migration networks and are positively correlated with the amount of remittances received (Borraz and others, 2005; Acosta, 2006, 2011; Salas, 2014). These migration networks provide the mechanism to transmit the effects of the unemployment shocks from these destination countries to the recipient households. More importantly for identification, this interaction allows for the introduction of the region fixed effects in our extensive margin estimates to address potential unobservables at the regional level. We also include the regional Gini coefficient and the unemployment rates for each year in the analysis to control for any region-specific trend not captured by the region FE.[12]

Instrumenting remittances with the interaction term is similar to a difference-in-difference estimation strategy where the treatment is a continuous variable. The first stage compares remittance-receiving households in regions with relative high migration whose migrants are affected by the business cycle of the migrants' destination country against households in regions with less migration who are less exposed to those negative shocks. The second stage estimates create a similar relationship but with child labor as the dependent variable.

The IV can therefore be expressed as:

$$RegionUnempShock_{rt} = \sum_{j=1}^{J} s_{rj} \times DestUnempShock_{jt}$$

where $DestUnempShock_{jt}$ is the deviation of the 12-month average unemployment rate (that ended

[11] Between November of 2007 and June 2009, the United States economy contracted by 5 percentage points and once the crisis spread to the Eurozone, the Spanish economy contracted by 3.6 percentage points between 2008 and 2010. Unemployment rates increased by 86 percent and 125 percent in the United States and Spain respectively between 2008 and 2010. During this period, the unemployment rate in Venezuela fluctuated between 7 and 10 percent.

[12] For example, these regional control variables address the potential threat to identification from the direct effect of shocks at a destination country to specific regions with stronger ties to that country through the regional labor market.

at time t) from the pre-crisis (2007) average unemployment rate in destination country j.[13] This destination country unemployment shock is then weighted with the historical share of Colombian migrants from region r that went to country j, s_{rj}. Data for the unemployment was obtained from Eurostat for Spain, the Bureau of Labor Statistics for the United States, and the National Institute of Statistics (INES) for Venezuela, while the historical share of Colombian migrants by region was calculated based on the IPUMS sample of the 2005 Census data (see Table 4).

To illustrate the relevance of this IV, we regress unemployment shocks on household receipts of international remittances for regions with high, moderate, and low shares of historical migrants to the United States and Spain and compare them with the remittance receipts of those living in regions with very low shares of historical migrants to these countries. We therefore estimate:

$$Remittances_{ht} = \alpha_0 + treat_I \times RegionUnempShock_{rt} + \mathbf{X_{it}}\beta + \psi_r + \varepsilon_{it}$$

where $I \in (H, M, L)$ and H, M, and L respectively indicate regions with high, moderate, and low shares of historical migrants – corresponding to regions at the top, second, and third quartiles of historical migrant shares to the United States and Spain. They are compared with those in the fourth quartile, which have less than 1 percent migrant share to these countries. ψ_r indicates region fixed effects. We present the results in Table 5. They show that the magnitude of the negative impact of the unemployment shocks on remittances monotonically increases with the share of migrants to these countries.

Finally, our identification relies on the assumption that the increase in unemployment rates in the United States and Spain was unexpected for Colombian households. Given sufficient time, households would employ multiple strategies to adjust to the unexpected decline in the remittance amount (Yang, 2008; Arango et al., 2015). To limit the impact of the medium- and long-term endogenous adjustments at the household level, we focus our analysis to the 2008–2010 period.

6 The Average Impact of Remittances on Child Labor

6.1 Extensive Margin: Labor Market Participation

Table 6 reports our results for the impact of remittances on child labor. All standard errors are clustered at the household level. The first three columns are estimates of the impact at the extensive margin. In Panel A, we report the standard LPM estimates. The estimates are statistically significant and indicate a negative relationship between remittances and child labor. In Column 1, we only include the region fixed effects. We then control for child, household, and household head characteristics in Column 2 and additional regional-level variables in Column 3. The estimated coefficients are qualitatively similar.

[13] This regional unemployment shock variable is analogous to the shock used in Yang (2008), but unlike his, our data only allow us to construct the measure at the regional (instead of the household) level.

Next, we implemented our 2SLS estimation using the interaction between the share of migrants at the main destinations and the unemployment rate on those places as the IV (see Section 5). Panel C in Table 6 reports the first-stage estimates. All columns indicate a negative correlation between the instrument and the remittances that households receive: a negative unemployment shock on migrants' destination is associated with receiving fewer remittances as expected. Given the somewhat low the Kleibergen-Papp F-statistic, we also report the Anderson-Rubin (AR) confidence interval as a test for the robustness of our estimates in the presence of weak instrument (Dufour and Taamouti, 2005). We find that the 90 percent AR confidence interval of our estimate does not include zero.

As the variation generated by the instrumental variable now also varies within each region, we are able to control for the potential effects of any time invariant unobserved region characteristic and improve identification by including the region fixed effects in our estimation (Hanson and Woodruff, 2003). Therefore, the first and second-stage equations now become:

$$Y_{iht} = \gamma R_{ht} + \mathbf{X_{it}}\beta + \psi_r + \varepsilon_{it} \tag{1}$$

$$R_{ht} = \alpha_1 RegionUnempShock_{rt} + \mathbf{X_{it}}\beta + \psi_r + \mu_{it} \tag{2}$$

where ψ_r are the region fixed effects.

The local average treatment effect (LATE) estimates from the 2SLS are reported in Columns 1–3 of Panel B. Column 3 presents the results from our preferred specification. With the full set of controls, a US$1 increase in remittances decreases the incidence of child labor by 0.017 p.p., an effect that is statistically significant at the five percent level. This suggests that a 10-percent increase at the mean remittance value of US$ 4,040 leads to a 6.8 p.p. reduction in the probability of child labor.[14]

6.2 Intensive Margin: Working Hours

Next, we study the impact of remittances on child labor at the intensive margin. In particular, we estimate the following IV Tobit model:

$$Y_{iht} = \alpha_0 + \mathbf{X_{it}}\beta + \gamma R^*_{ht} + \mu_{it} \tag{3}$$

$$R^*_{ht} = \mathbf{X_{it}}\Pi_1 + RegionUnempShock_{rt}\Pi_2 + \eta_{it} \tag{4}$$

where α, β, and γ are vectors of structural parameters. R_{ht} is the total remittances received by the child's household, Π_1 is a matrix of reduced-form parameters, and Π_2 is the parameter of the instrumental variable, $RegionUnempShock_{rt}$. The estimated IV-Tobit model therefore accounts for

[14]The magnitude of this effect on child labor is smaller compared to the effect estimated in Calero et al. (2009) for Ecuador. However, in their study, the effect of remittance on child labor is not statistically significant.

the censored nature of the dependent variable and the (continuous) endogenous regressor. The model is estimated using the Newey (1987) two-step estimator with bootstrapped standard errors.

As in the case of our extensive margin estimation, we would like to introduce the region fixed effects in this specification. However, including these fixed effects in our nonlinear Tobit model will introduce the incidental parameters problem. Greene (2004) shows that with Tobit estimates, the incidental parameter problems will bias estimates of the coefficients and the standard errors, and the problem is exacerbated by the degree of censoring in the data. We therefore do not include the region fixed effects in our preferred specification, although we include such an estimate in the appendix as a robustness check.[15]

We report the marginal effects of remittance on child labor in Column 4 of Table 6. The F-test for the first stage suggests that the IV is relevant for our specification.[16] We find that remittances reduce child labor at the intensive margin. The effect of remittance income on hours worked is negative and statistically significant. The marginal effect estimate suggests that a 10 percent increase at the mean of remittances leads to around 2.3 hours reduction (or about 6 percent reduction) in the number of hours worked during the week.

7 Heterogeneous Impacts: Demographic and Economic Factors

Economic and institutional factors can lead to the heterogeneous treatment impacts by gender and age. Traditional gender roles in economic activities can lead to heterogeneous impacts of remittances by gender (Edmonds, 2007).[17] In terms of age, theoretically parental (and governmental) preference to keep younger children out of work, as well as higher labor market returns for older children, will increase the reservation wage for younger children. However, with participation that increases in age, perhaps there is a greater margin for improvement among older children. Overall, the age impact is empirically ambiguous.

Furthermore, as we show in the conceptual framework, the decision of send children to work depends on the household reservation wage. Children are sent to work when the market wage is higher than the reservation wage. For poor households, the remittance income may not be enough to offset their children's income. Hence, to the extent that richer households have higher reservation wages, the negative effect of remittances on the children-labor supply of richer households will be stronger. At the same time, there may be a greater margin for improvement among poor households since children in those households are more likely to work.

In this section, we empirically examine these heterogeneous impacts. To study these impacts,

[15] Table A.3 presents the estimate of the IV-Tobit model with the region fixed effects. The main effect is smaller but remains statistically significant.

[16] In the appendix, we estimate a 2SLS model of hours worked as a comparison. The results are presented in Table A.4. Under this specification, the Kleibergen-Paap statistic suggests that we have a strong IV.

[17] In Colombia, for example, Table 1 shows that a higher proportion of girls reported being employed in the domestic sector.

we disaggregate the average impact of remittances by demographic and economic factors by estimating the remittance impacts by each subgroup. We explore gender and age effects, as well as the effect of household wealth, as proxied by the aforementioned asset-based wealth index.

7.1 Demographic Heterogeneity: Gender and Age

Before we report our results on the disaggregated impacts by demographic characteristics, we first examine whether there is selection into the remittance-receiving households by these characteristics. Table A.2 in the Appendix compares the share of male and the mean age of children by whether the respondent lives in a remittance-receiving (or non-remittance-receiving) household. For each variable, we conducted a *t-test* of whether this variable differs between the two types of households. We do not find any statistically significant difference in the gender or age composition of children between the two types of households.

Table 7 presents the results for our analysis of the treatment effects that is disaggregated by gender and age. We obtain these results by estimating the models for the relevant sub-sample. Column 1 reports the results from our baseline specification for comparison.

Gender Effects. — Columns 2–3 report the effects of remittances by gender. Although the Kleibergen-Paap statistics are relatively low for both sub-samples, the AR test suggests that the estimates are robust to the presence of weak instruments. Panel A reports that at the extensive margin, the impact is only significant for girls. However, the extensive margin estimates are not statistically different between the estimate for girls and boys. At the intensive margin, although the magnitude of the coefficient is larger for boys, the estimate for their sample subset is imprecisely estimated.

Previous evidence suggests, on the one hand, that remittance income reduces labor supply among both boys and girls (Acosta, 2011; Bargain and Boutin, 2015). On the other hand, evidence from government transfer programs suggests a stronger effect for boys, especially when the emphasis is on market work (de Hoop and Rosati, 2014a). Our results for Colombia, where the estimates are only significant among girls, may be explained by the greater social acceptability of work for boys compared to girls in Latin America (reflected in their average working hours, see Table 2). The idea that boys are expected to work more than girls would increase the reservation wage for girls and may explain why the effects are only statistically significant for girls (Brown et al., 2003; Duryea and Arends-Kuenning, 2003).

Age Effects. — To study the heterogeneous impact of remittances by age groups, we divide the age groups into two: Younger children aged 12–14 years old, and older children aged 15–18 years old. These age groups mimic those stipulated in the law on child labor (see Section 2), except that we conflate the two older age groups.[18]

We report the heterogeneous treatment effects by age groups in Columns 4–5 of Table 7.

[18] We estimated the effect for the three age groups stipulated in the law. We found that the effects are less precisely estimated among the older children. These results are available from the authors.

Overall, the differential impact of remittances on child labor between the two age groups is small. However, in terms of hours worked, the effect is only statistically significant for younger children. At the extensive margin, we find that the difference in the point estimates between the two age groups is not statistically different. The coefficient for older children is statistically significant, but only at the 10 percent level.[19] Meanwhile, at the intensive margin, the remittance effect is only statistically significant (at the 5 percent level) for younger children.

As such, we do not find strong evidence for age-differential effects of remittances on the likelihood to work, and some evidence of a stronger effect on work hours for younger children. This finding is consistent with the existing literature that remittances tend to reduce child labor among younger children (Acosta, 2011; Bargain and Boutin, 2015). It is also consistent with the interpretation that parents tend to assign a higher reservation wage to younger children. Our finding contrasts with the evidence on the impact of conditional cash transfers on child labor, which finds little age-differential impact (de Hoop and Rosati, 2014a).

7.2 Wealth Heterogeneity

To study the role of household economic status, we disaggregate the sample into poor and non-poor households, based on the constructed wealth index (see Section 4.2). We define as poor (non-poor) households whose wealth index is below (above) the median wealth index. Columns 7 and 8 in Table 7 examine the heterogeneous impacts of remittances between poor and non-poor households.

We find that the impact is negative for both household types, but only significant for children in non-poor households. At the extensive margin, we do not find statistically significant differences in the magnitude of the coefficients among both types of households, and the coefficient for non-poor households is only significant at the 10 percent level. At the intensive margin, however, this effect is only precisely estimated for non-poor households. These findings suggest a higher reservation wage for non-poor households, as is the case in Ecuador (Calero et al., 2009). However, the result for poor households needs to be interpreted with caution given the weak instrument for this sub-sample.

8 Remittances and Schooling

In many developing countries, the primary opportunity costs from school include the returns from labor market (Rosenzweig, 1990). So far, our findings suggest that remittance income is effective overall in reducing child labor participation and hours worked. There is evidence from some other countries in Latin America that remittances may also have the capacity to increase schooling (Calero et al., 2009; Amuedo-Dorantes and Pozo, 2010; Alcaraz et al., 2012), although the

[19] The standard errors in the estimates for younger children are larger, in part due to their relatively smaller sample size compared to those for the older cohort.

effect is by no means universal (see e.g., Acosta, 2011). In this section, we provide the empirical evidence on the impact of remittance on schooling in Colombia.

We do not find evidence that remittances increase schooling in Colombia. GEIH collects school attendance information from the children of the households. Using the same identification strategy for the extensive margin above, we estimate the impact of remittance on the likelihood of attending school. Table 8 reports our results. Columns 1 and 2 report standard LPM estimates without and with individual and household control variables, while Column 3 reports the 2SLS estimate. All standard errors are clustered at the household level. Despite positive and statistically significant correlation between remittance and schooling in Column 1, the correlation disappears with the full set of controls and the instrumental variable estimation (Columns 2 and 3).

More broadly, this finding illustrates the heterogeneity of the the mechanisms through which remittance influences the welfare of children left behind in the region. On the one hand, our finding contrasts with that of Calero et al. (2009), who found a statistically significant effect of remittances on school attendance, but not on child labor in Ecuador (which borders with Colombia). On the other hand, Acosta (2011) similarly finds that remittance does not affect schooling in El Salvador. In the case of Colombia, the evidence supports the notion that the negative link between child labor and schooling may be weak, perhaps because most working children also attend school. (Edmonds and Pavcnik, 2005).[20]

9 Conclusion

Remittance income provides an important means to relax liquidity constraints for households. By relaxing budget constraints, it reduces child participation in the labor market that may have negative long term consequences. We show that remittances reduce child labor in Colombia. Given concerns that parental absence due to migration can increase pressure for children to work to compensate for lost income (Antman, 2012; Giannelli and Mangiavacchi, 2010), we show that remittance income can mitigate the potential negative impacts of migration on the welfare of children who are left behind.

Our findings also highlight the importance of collecting a richer set of child labor outcomes for a more complete pictures of the impact of policy changes on child labor. For example, with some recent exceptions, the literature on the impact of remittances on child labor primarily studies children's participation instead of hours worked. Without altering employment status, remittances can reduce hours worked and the exposure to labor for children left behind. We find that without data on hours worked, we would not be able to capture some of the heterogeneous impact of remittances across different groups. At the same time, having information on migration status can help to improve identification in this study.

We propose two avenues for future research. First, there is the question of the role of intra-

[20] In our sample, 80 percent of working children spend most of their time studying.

household bargaining as a mechanism that can help explain the heterogeneous results found in this paper. The absence of a household member and the imperfect monitoring by migrating adults can shift the intra-household allocation of resources (including the remittance income) among the household members left behind (Chen, 2006; Mendola and Carletto, 2012). Variation in the gender of the household member who migrated can affect the welfare outcomes of children left behind by influencing intra-household allocations. Second, it would be important to extend the analysis in this paper to children living in hard-to-reach populations. This is particularly relevant for a country like Colombia where internal conflicts have led many children to engage in some of the worst forms of child labor.

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Tables

Table 1: Child Labor by Economic Activity

	All	Boys	Girls	12-14	15-16	17-18
Wholesale and retail	27.59	26.14	29.61	37.66	28.24	24.33
Hotels and restaurants	9.05	7.61	11.05	9.71	8.98	8.88
Manufacturing	7.26	6.27	8.65	12.92	7.25	5.60
Storage transportation and communication	6.72	5.46	8.49	9.74	6.95	5.74
Other services	2.65	3.05	2.09	2.32	2.94	2.61
Construction	5.32	8.93	0.25	1.89	5.19	6.55
Domestic service	5.60	0.29	13.03	2.09	5.49	6.68
Others	35.81	42.25	26.83	23.67	34.96	39.61
All sectors	100	100	100	100	100	100

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010.* We use the 2-digit International Standard Industrial Classification (ISIC) to classify the economic activities. We present the five most important economic sectors, in addition to the construction and domestic services sector. None of the omitted activities (aggregated as "Others") accounts for more than five percent of working children.

Table 2: Average Working Hours during the Week

	All	Age groups		
	12-18	12-14	15-16	17-18
All	34.95	21.80	31.75	40.26
Boys	36.06	23.13	32.85	41.43
Girls	33.39	19.99	30.04	38.68
N	17,926	3,007	4,662	10,257

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010.*

Table 3: Descriptive Statistics

	Recipient	Non-recipient
Outcome variables		
Labor Participation	10.45	12.30
Hours worked	31.46	35.07
Children Characteristics		
Male	49.66	49.98
Age	15.07	15.03
Household characteristics		
Other children in HH	2.39	2.48
Household members	4.98	5.03
Wealth index	1.90	1.59
Household head characteristics		
Years of education	8.24	7.98
Household head is [...]		
Female	56.63	36.25
Employed	58.17	77.64
Married	32.24	36.15
Total remittances amount	4,040	0
N	5,713	141,060

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. Total amount of remittances in PPP-adjusted US dollar.

Table 4: Share of migrants by region and destination

	Total	Share of population who migrated to [. . .]			
	population	All destinations	United States	Spain	Venezuela
Pereira	572,860	17.92	7.98	7.64	2.30
Cúcuta	663,270	16.23	0.00	0.22	16.01
Cali	2,560,721	10.38	4.05	3.08	3.26
Bucaramanga	915,531	8.71	1.19	0.71	6.81
Villavicencio	386,988	7.97	1.83	3.26	2.88
Medellín	3,046,541	7.94	4.80	1.49	1.66
Bogotá	7,029,970	7.85	3.60	1.74	2.50
Ibagué	431,005	5.80	1.66	0.69	3.45
Barranquilla	1,523,223	4.81	1.24	0.63	2.93
Cartagena	974,463	4.40	0.06	0.20	4.14
Manizales	454,228	3.08	2.05	0.94	0.10
Montería	522,382	1.54	0.32	0.12	1.09
Pasto	347,655	0.15	0.00	0.00	0.15

Notes: We use the question on the respondent's country of residence five years ago from IPUMS to compute total number of migrants and construct the distribution of migrants by metropolitan area. We then use the distribution of migrants and the number of Colombian migrants in the United States, Spain and Venezuela to construct the share of migrants. Data of Colombians migrants in the US comes from the Pew Hispanic Research Center; in Spain from *Ministerio de Empleo y Seguridad Social de España*; and in Venezuela from *Instituto Nacional de Estadística*.

Table 5: Impact of unemployment shocks on remittances

	(1)	(2)	(3)
$Treat_H$	-0.094	-0.085	
	(0.035)***	(0.035)**	
$Treat_M$	-0.012	-0.009	
	(0.031)	(0.031)	
$Treat_L$	-0.003		0.043
	(0.080)		(0.082)
Region FE	Yes	Yes	Yes
Child characteristics	Yes	Yes	Yes
Household characteristics	Yes	Yes	Yes
Region Variables	Yes	Yes	Yes
Observations	146,773	103,279	86,615

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. The individual and household controls included are presented in Table 3. Regional controls include the Gini coefficient and the unemployment rate. Coefficients are reported with standard errors clustered at the household level in parentheses. */**/*** denotes significance at the 10/5/1 percent level.

Table 6: Remittances and Child Labor

	Likelihood to work (Linear probability model)			Hours (Tobit)
	(1)	(2)	(3)	(4)
Panel A: Non-IV Estimates				
Remittances	-0.0003	-0.0001	-0.0001	-0.0103
	(0.0001)***	(0.0001)**	(0.0001)**	(0.0053)**
Panel B: IV Estimates				
Remittances	-0.011	-0.009	-0.017	-0.558
	(0.005)**	(0.005)*	(0.008)**	(0.157)***
Dependent variable (panel C):	Remittances			
Panel C: First Stage				
RegionUnempShock	-0.084	-0.090	-0.073	-0.092
	(0.030)***	(0.030)***	(0.027)***	(0.025)***
Kleibergen-Paap rk Wald F statistic	7.99	8.98	7.39	
F-statistic				14.10
Anderson-Rubin Test	[-0.03,-0.01]	[-0.02,-0.003]	[-0.04,-0.01]	
Region FE	Yes	Yes	Yes	No
Child characteristics	No	Yes	Yes	Yes
Household characteristics	No	Yes	Yes	Yes
Region Variables	No	No	Yes	Yes
Observations	146,773	146,773	146,773	146,773

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. The individual and household controls included are presented in Table 3. Regional controls include the Gini coefficient, the unemployment rate, and the historical net migration between 2000-2005 for the regions. Columns 1-3 in Panel B report coefficients from the LPM with standard errors clustered at the household level in parentheses. Column 4 reports the marginal effect from the IV-Tobit calculated based on Wooldridge (2010, Section 17.2) and the standard error is bootstrapped with 500 replications. The point estimates and standard errors for the LPM, and the marginal effect and standard error for the Tobit model in Panel A and Panel B are multiplied by 100 for presentation purposes. We report Anderson-Rubin 90 percent confidence intervals. */**/*** denotes significance at the 10/5/1 percent level.

Table 7: Heterogeneous Treatment Effect of Remittances

		Gender		Age		Wealth	
	All	Boys	Girls	12-14	15-18	Poor	Non-poor
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Likelihood to work							
Remittances	-0.017	-0.015	-0.019	-0.015	-0.020	-0.098	-0.011
	(0.008)**	(0.011)	(0.010)**	(0.010)	(0.011)*	(0.190)	(0.006)*
Kleibergen-Paap F-statistic	7.39	3.09	5.98	3.02	6.03	0.28	5.74
Anderson-Rubin Test	[-0.04,-0.01]	[-0.05,-0.002]	[-0.05,-0.01]	[-0.05, -0.01]	[-0.06,-0.01]	[-0.72, 0.53]	[-0.03, -0.004]
Panel B: Hours worked							
Remittances	-0.558	-0.634	-0.538	-0.468	-0.470	-2.478	-0.337
	(0.157)***	(1.321)	(0.229)**	(0.184)**	(0.416)	(12.392)	(0.159)**
F-statistic	14.10	5.79	11.06	6.61	10.69	2.01	9.62
Observations	146,773	73,340	73,433	61,821	84,952	79,382	67,391

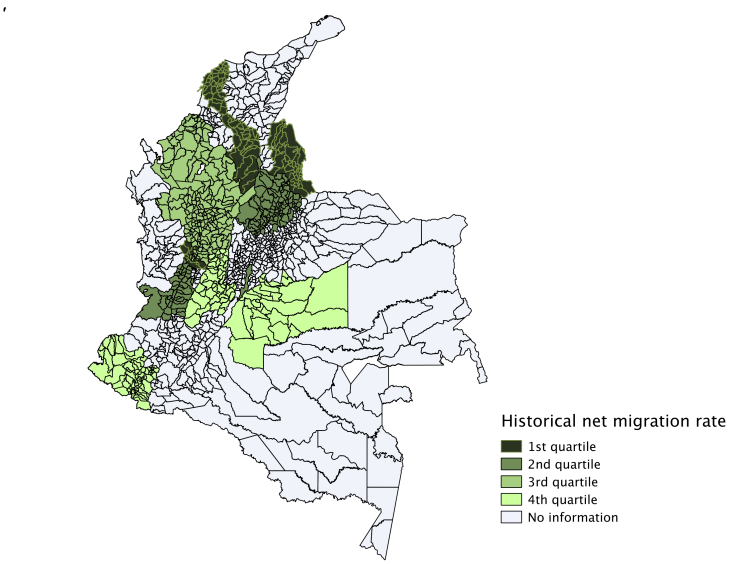
Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. The controls included are the same as in Table 6. Panel A reports coefficients from the LPM with standard errors clustered at the household level in parentheses. Panel B reports marginal effects from the IV-Tobit calculated based on Wooldridge (2010, Section 17.2) and the standard errors are bootstrapped with 500 replications. The point estimates and standard errors for the LPM and the marginal effects and standard errors for the IV-Tobit model are multiplied by 100 for presentation purposes. We report Anderson-Rubin 90 percent confidence intervals. */**/*** denotes significance at the 10/5/1 percent level.

Table 8: Remittances and school attendance

	School attendance		
	OLS	OLS	2SLS
Remittances	0.0003	0.0001	-0.0089
	(0.0001)***	(0.0001)	(0.0064)
Kleibergen-Paap rk Wald F statistic			7.39
Anderson-Rubin Test			[-0.03,-0.0001]
Observations	146,773	146,773	146,773
Region FE	Yes	Yes	Yes
Child characteristics	No	Yes	Yes
Household characteristics	No	Yes	Yes
Region Variables	No	Yes	Yes

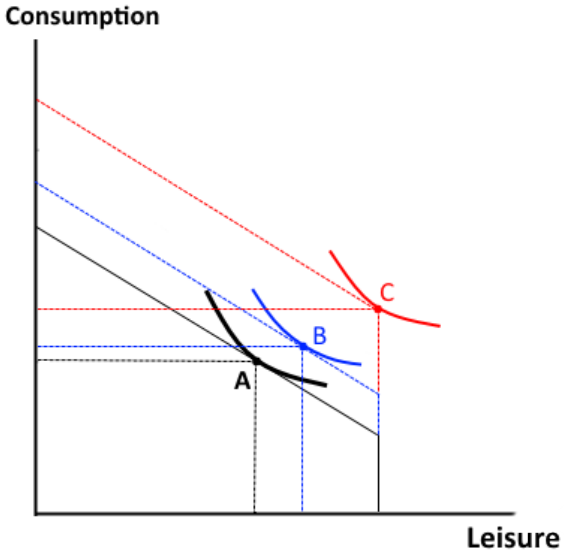
Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. The individual and household controls included are presented in Table 3. Regional controls include the Gini coefficient and the unemployment rate. Columns 1-3 report coefficients from the LPM with standard errors clustered at the household level in parentheses. The point estimates and standard errors for the LPM are multiplied by 100. We report Anderson-Rubin 90 percent confidence intervals. */**/*** denotes significance at the 10/5/1 percent level.

Figure 1: Historical Net Migration Rate



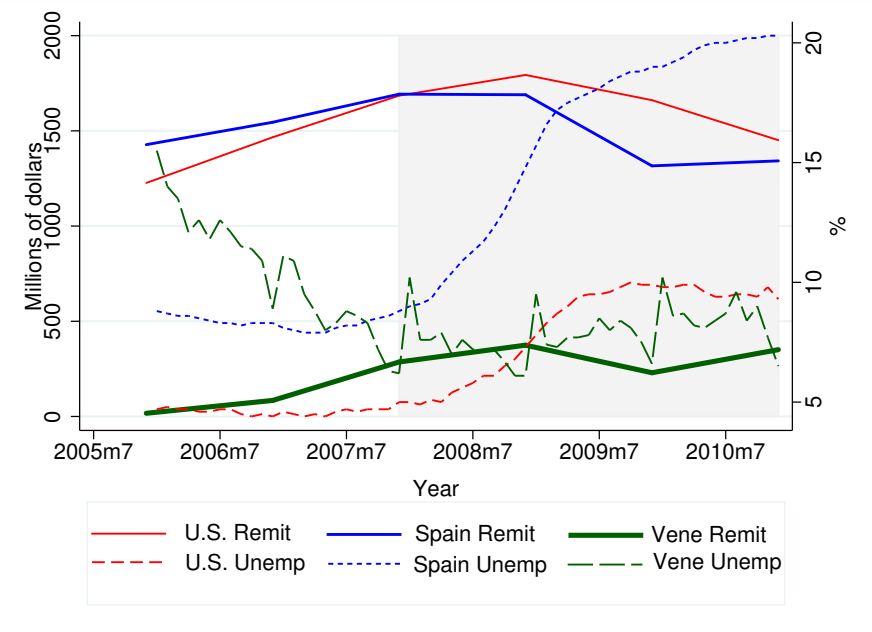
Notes: Historical net migration rate is measured as immigrants minus migrants in the numerator with respect to the average initial population plus the final population during 2000-2005. Net migration rate figures are calculated based on data from DANE.

Figure 2: The Impact of Remittances on Child Labor



Source: Authors

FIGURE 3: REMITTANCE FLOWS AND DESTINATION COUNTRY UNEMPLOYMENT RATES



Notes: Remittance flows are obtained from the Colombian central bank, *Banco de la República*, and unemployment rates are obtained from Eurostat for Spain, the Bureau of Labor Statistics for the United States, and the National Institute of Statistics (INES) for Venezuela.

Appendix

Section A.1 describes the principal component analysis and the list of variables used to construct the wealth index. Section A.2 reports additional tables mentioned in the paper.

A.1 Wealth Index: Principal Component Analysis

Our wealth index is constructed based on a set of household assets using a principal component analysis (Vyas and Kumaranayake, 2006). In particular, given the vector of assets X, the first principal component is the linear combination:

$$Y = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n$$

where Y is our principal component index that proxies for household wealth, x_n is a standardized variable with zero mean, and a_n are the coefficient estimates. The principal component reports an index which assigns a larger weight to assets which vary most among households. Given that we use dummy variables, the coefficients capture the effects of a change from zero to one in each particular asset on the wealth index. We use the variable Y as the index to classify households into different socio-economic groups. Table A.1 presents the full list of the variables that are used as inputs to our wealth index and their respective weights.

Table A.1: Results from the Principal Components Analysis

	Mean	Std.dev.	Factor score
Home ownership	0.435	0.496	0.087
Electricity	0.959	0.199	0.199
Television	0.617	0.486	0.264
Refrigerator	0.841	0.366	0.272
Car	0.159	0.365	0.200
Telephone	0.572	0.495	0.309
Internet	0.276	0.447	0.219
Source of water supply			
Aqueduct pipe	0.961	0.193	0.299
Pipe to other source	0.008	0.090	-0.155
Well with pump	0.002	0.047	-0.050
Water tank	0.002	0.040	-0.049
Storm water	0.0002	0.013	-0.022
River	0.001	0.021	-0.034
Public stack	0.002	0 .041	-0.079
Trolley tank	0.0002	0.014	-0.023
Water bearer	0.004	0.059	-0.131
Bottled water	0.005	0.070	-0.053
Sanitation facility			
Toilet connected to [. . .]			
sewage	0.903	0.295	0.380
septic tank	0.065	0.247	-0.268
river or sea	0.0003	0.017	-0.137
Toilet off line	0.010	0.097	-0.047
Latrine	0.001	0.028	-0.046
No service	0.006	0.074	-0.201
Type of floor material			
Ground/sand	0.023	0.148	-0.201
Cement	0.263	0.440	-0.246
Rough wood	0.025	0.156	-0.011
Brick	0.620	0.485	0.351
Marble	0.008	0.087	0.032
Polished wood	0.014	0.118	0.025
Carpet	0.006	0.076	0.026

Notes: The sample includes all households from *GEIH, 2008-2010*.

A.2 Additional Tables

Table A.2: Socio-demographic Characteristics of Remittance Recipients and Non-recipients

	Recipient	Non-recipient	$P(\|\mu_r - \mu_{nr}\| \neq 0)$
Male	0.497	0.499	0.634
Age	15.02	15.07	0.155

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*.

Table A.3: The Heterogeneous Effect of Remittances on Hours Worked using Region FE

		Gender		Age		Wealth	
	All	Boys	Girls	12-14	15-18	Poor	Non-poor
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Remittances	-0.362	-0.387	-0.366	-0.453	-0.247	-2.131	-0.318
	(0.201)*	(7.200)	(2.557)	(1.728)	(1.467)	(19.143)	(0.199)
Observations	146,773	73,340	73,433	61,821	84,952	79,382	67,391

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. The controls included are the same as in Column 4 of Table 6, but with region fixed effects. We report marginal effects from the IV-Tobit calculated based on Wooldridge (2010, Section 17.2) and the standard errors are bootstrapped with 500 replications. The marginal effects and standard errors are multiplied by 100 for presentation purposes. */**/*** denotes significance at the 10/5/1 percent level.

Table A.4: Heterogenous Effects of Remittances with Two-Stage Least Squares

		Gender		Age		Wealth	
	All	Boys	Girls	12-14	15-18	Poor	Non-poor
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Remittances	-0.667	-0.558	-0.659	-0.181	-1.152	-3.037	-0.304
	(0.209)***	(0.325)*	(0.232)***	(0.105)*	(0.398)***	(2.135)	(0.150)**
Kleibergen-Paap rk F statistic:	15.60	5.79	12.39	7.28	11.40	2.20	10.01
Anderson-Rubin Test	[-0.98, -0.27]	[-1.63, -0.18]	[-1.14, -0.27]	[-0.50, -0.18]	[-1.82, -0.45]	[-10.06, -1.26]	[-0.48, 0.03]
Observations	146,773	73,340	73,433	61,821	84,952	79,382	67,391

Notes: The sample includes children between 12-18 years old from *GEIH, 2008-2010*. The controls included are the same as in Column 4 of Table 6. The estimates and standard errors are multiplied by 100 for presentation purposes. We report Anderson-Rubin 90 percent confidence intervals. */**/*** denotes significance at the 10/5/1 percent level.